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Editorial

Vasopressors or fluids for initial resuscitation in septic shock: A matter of clinical judgement

Septic shock is characterised clinically by the requirement for vasopressors to maintain a mean arterial blood pressure (MAP) of 65 mmHg and a blood lactate >2 mmol/L that persists after fluid resuscitation.[1] The question of the optimal volume of intravenous (IV) fluid resuscitation and the appropriate clinical and physiological targets has been debated over the past two decades. A landmark randomised trial of a multifaceted intervention in adults with septic shock introduced the concept of early goal-directed therapy (EGDT).[2] While subsequent trials found no benefit from the overall EGDT 'bundle' in patients with early septic shock, an initial IV resuscitation fluid volume of 3–4 L has been adopted into usual practice.[3] The international Surviving Sepsis Campaign guidelines also provide a weak recommendation for an initial IV fluid volume of at least 30 mL/kg of isotonic crystalloid within the first 3 h, based on a paucity of high-quality evidence.[4] This recommendation aligns with recent guidance from the European Society of Intensive Care Medicine for initial haemodynamic resuscitation in adults with septic shock.[5] Meanwhile, observational data have fostered emerging concerns about harm from excess IV fluid and a positive cumulative fluid balance in critically ill patients with septic shock.[6–8]

The second element of resuscitation in septic shock is the timing of introduction of a vasopressor infusion. Observational studies and two small randomised trials of fixed, low-dose norepinephrine have suggested a benefit with earlier vasopressor introduction.[9] Notably, the 2026 Surviving Sepsis Campaign guidelines make no specific recommendation regarding the timing of vasopressor initiation, other than that concurrent vasopressor and fluid administration may be warranted in unstable septic shock.[4]

The ARISE FLUIDS open-label, randomised trial provides important new evidence to inform this debate.[10] Patients with septic shock presenting to the emergency department (ED) were randomised to a strategy of early vasopressors and restricted IV fluids, or a larger initial IV fluid volume followed by a vasopressor infusion if required. There was no difference in the primary outcome of days alive and out of hospital at day 90 (DAOH90) post-randomisation overall, nor in any prespecified subgroups. There was no difference in the secondary outcome of 90-day all-cause mortality. Both strategies appeared safe, although pulmonary oedema was reported more frequently in the group that received a larger fluid volume.

ARISE FLUIDS differs from recent randomised trials investigating a restricted fluid strategy in septic shock. The CLASSIC trial randomised 1554 patients with septic shock in the intensive care unit (ICU) to a restricted IV fluid regimen for the whole of the ICU stay, compared to usual care.[11] While no difference in 90-day mortality was observed, participants in both groups had already received >3 L of IV fluid and were receiving a vasopressor infusion pre-randomisation. The CLOVERS trial, which recruited predominantly in the ED, sought to investigate the initial resuscitation phase. Patients with sepsis and a systolic blood pressure (SBP) < 100 mmHg were randomised to liberal IV fluid management or a restricted fluid and early vasopressor strategy for 24 h.[12] The trial was stopped for futility following a planned interim analysis at two-thirds recruitment. Among the 1563 participants analysed, in-hospital mortality censored at 90 days was similar between groups. Participants in both groups received a median of >2 L of IV fluids pre-randomisation (≈30 mL/kg).

ARISE FLUIDS, conducted at 51 hospitals across Australia, New Zealand and the Republic of Ireland, recruited 1000 patients presenting to the ED with septic shock, defined as SBP <90 mmHg or MAP <65 mmHg after >1 L of bolus IV fluid, and blood lactate >2.0 mmol/L. Exclusion criteria included >2 L bolus IV fluid, a clinician-determined indication for, or contraindication to, IV fluid (e.g. dehydration, severe cardiac failure), or inability to deliver the trial protocol, such as operating theatre transfer or a treatment limitation that precluded vasopressor administration and/or ICU admission. Participants in both groups received a median IV fluid volume of 18 mL/kg pre-randomisation. Over the 24-h intervention period, participants in the vasopressor group received, ≈50% less IV fluid than the fluids group (median 14 mL/kg vs 29 mL/kg). A larger proportion of participants in the vasopressor group received a vasopressor infusion (86.5% vs 67.6%), which was started ≈1 h earlier than in the fluids group (median 21 min vs 81 min). Although there was a higher rate of ICU admission among the early vasopressor group in the first 24 h, the time to vasopressor cessation and time to ICU discharge were shorter. Vasopressors were initiated via a peripheral IV cannula in both groups with similar duration of peripheral administration.

The ARISE FLUIDS primary outcome (DAOH90) is a composite of death and days out of hospital up to 90 days post-randomisation, where death was assigned zero days. It is a consumer-informed outcome measure that aims to capture not just mortality but also the quality of survival.[13] Survival, disability and health-related quality of life at 12 months will be reported separately, along with a health economic analysis.

What are the implications for practice? ARISE FLUIDS is a pragmatic trial conducted across both large teaching hospitals and regional hospitals. Clinicians were permitted to exercise judgement over the precise fluid volume and vasopressor timing within the allocated group intervention. Resuscitation targets, including MAP, were clinician-determined, and measures to assess volume

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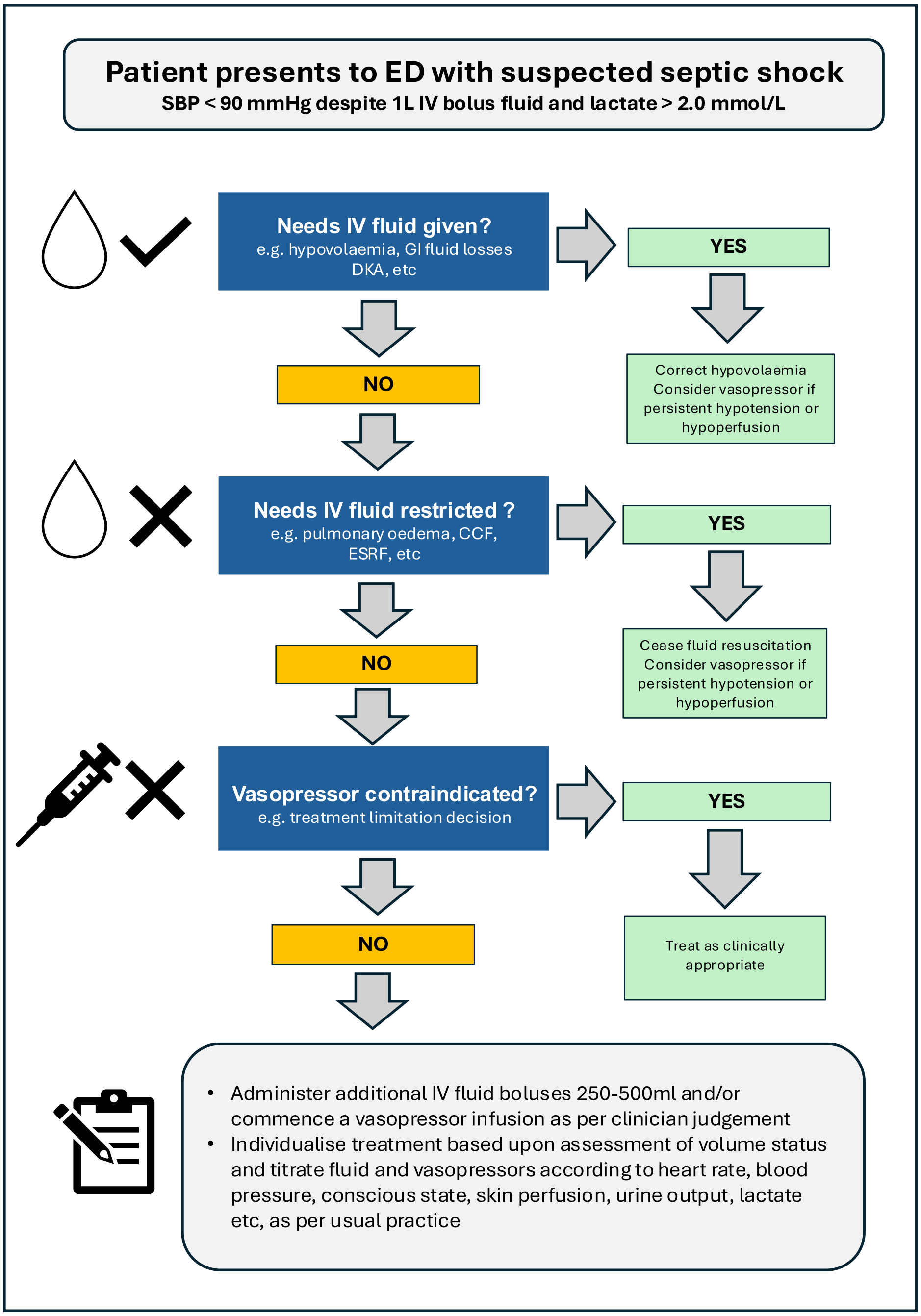

Fig. 1. Suggested approach to resuscitation for patients in the ED with suspected septic shock based upon the findings of ARISE FLUIDS. ED denotes emergency department; SBP, systolic blood pressure; IV, intravenous; GI, gastrointestinal; DKA, diabetic ketoacidosis; CCF, congestive cardiac failure; ESRF, end-stage renal failure.

status and fluid responsiveness were not specified. These factors provide confidence in translating the findings into everyday practice.

For patients fulfilling the trial enrolment criteria, ARISE FLUIDS does not support a specific fluid volume target for early haemodynamic resuscitation. Rather, IV fluids should be administered in aliquots titrated against bedside indicators of perfusion (Fig. 1). Similarly, the timing of vasopressor initiation is a matter of clinical judgment and may occur concurrently with fluid resuscitation. The low rates of complications seen in the trial provide reassurance about the safety of peripherally administered vasopressors in the initial management of septic shock. While both an early vasopressor and restricted fluid strategy and a larger fluid volume and later vasopressors resulted in similar outcomes, the organisational and resource implications for health systems require further evaluation.

Conflict of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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